

E-Commerce Database Design using SQL

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Project Description:

This project focuses on designing and implementing an E-Commerce database using SQL Server. It includes multiple related tables such as customers, products, orders, and payments. The project demonstrates relational database concepts using primary and foreign keys.

Various SQL queries were used to analyze business data, including total revenue calculation, customer purchase behavior, product performance, and payment analysis.

Database Design:

Tables Used:

- customers
- products
- orders
- payments

ER Diagram Screenshot:

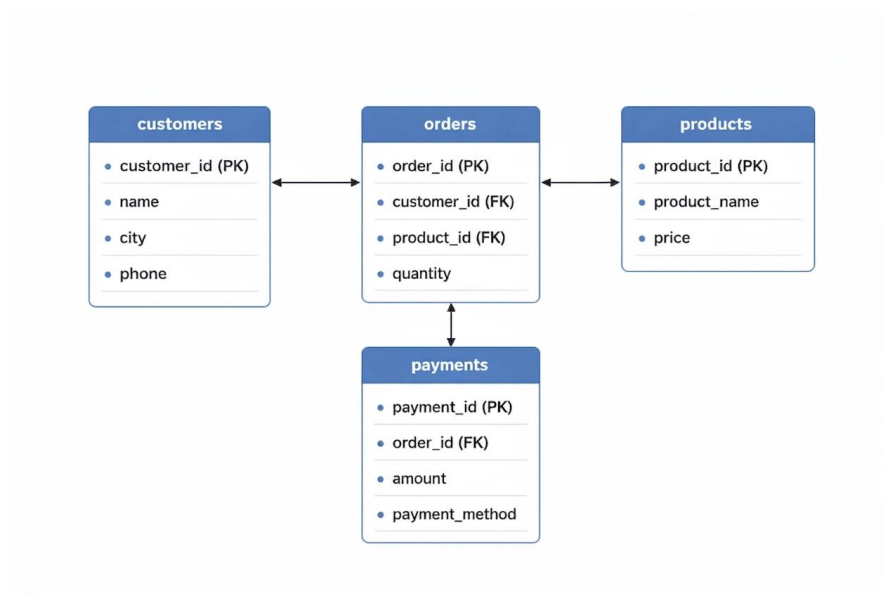
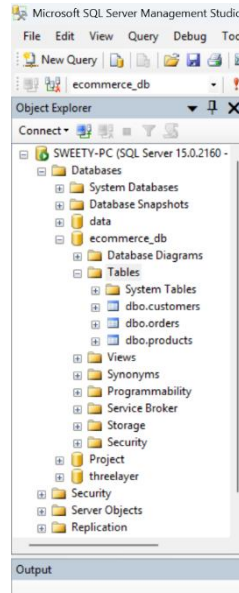
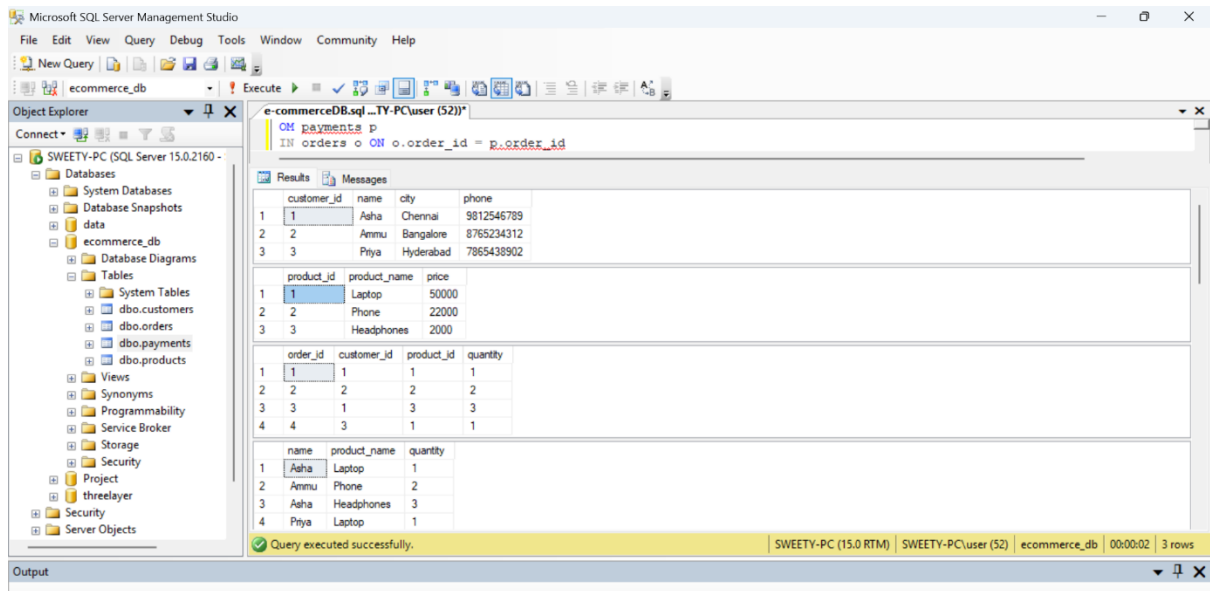


Table Structure:

Tables (customers, products, orders, payments)



Sample Data:



SQL Queries:

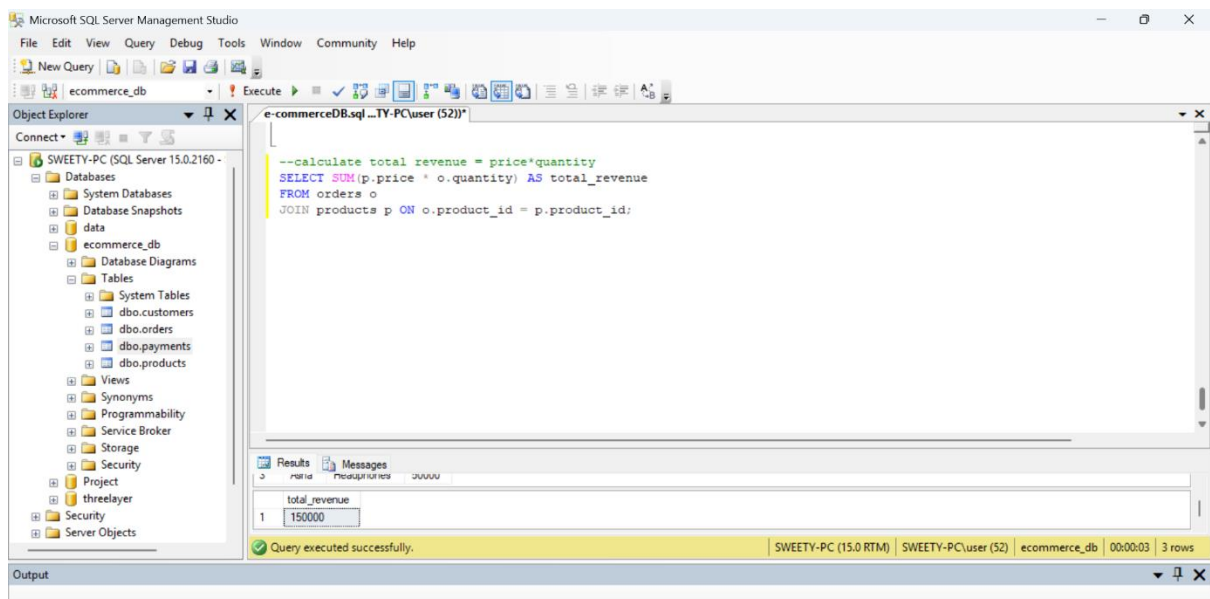
--calculate total revenue = price*quantity

```
SELECT SUM(p.price * o.quantity) AS total_revenue
```

```
FROM orders o
```

```
JOIN products p ON o.product_id = p.product_id;
```

Output:



--find top customer by total order

```
SELECT c.name, SUM(o.quantity) AS total_orders
```

```
FROM orders o
```

```
JOIN customers c ON o.customer_id = c.customer_id
```

```
GROUP BY c.name
```

```
ORDER BY total_orders DESC;
```

Output:

The screenshot shows the Microsoft SQL Server Management Studio interface. The Object Explorer on the left displays the database structure for 'ecommerce_db', including tables like 'customers', 'orders', 'payments', and 'products'. The main query window contains the following SQL code:

```
--find top customer by total order
SELECT c.name, SUM(o.quantity) AS total_orders
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY c.name
ORDER BY total_orders DESC;
```

The Results pane at the bottom shows the output of the query:

	name	total_orders
1	Asha	4
2	Ammu	2
3	Piya	1

The status bar at the bottom indicates the query was executed successfully on the 'ecommerce_db' database.

--most sold products

SELECT p.product_name, SUM(o.quantity) AS total_sold

FROM orders o

JOIN products p ON o.product_id = p.product_id

GROUP BY p.product_name

ORDER BY total_sold DESC;

Output:

The screenshot shows the Microsoft SQL Server Management Studio interface. The Object Explorer on the left displays the database structure for 'ecommerce_db'. The main query window contains the following SQL code:

```
--most sold products
SELECT p.product_name, SUM(o.quantity) AS total_sold
FROM orders o
JOIN products p ON o.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_sold DESC;
```

The Results pane at the bottom shows the output of the query:

	product_name	total_sold
1	Headphones	3
2	Laptop	2
3	Phone	2

The status bar at the bottom indicates the query was executed successfully on the 'ecommerce_db' database.

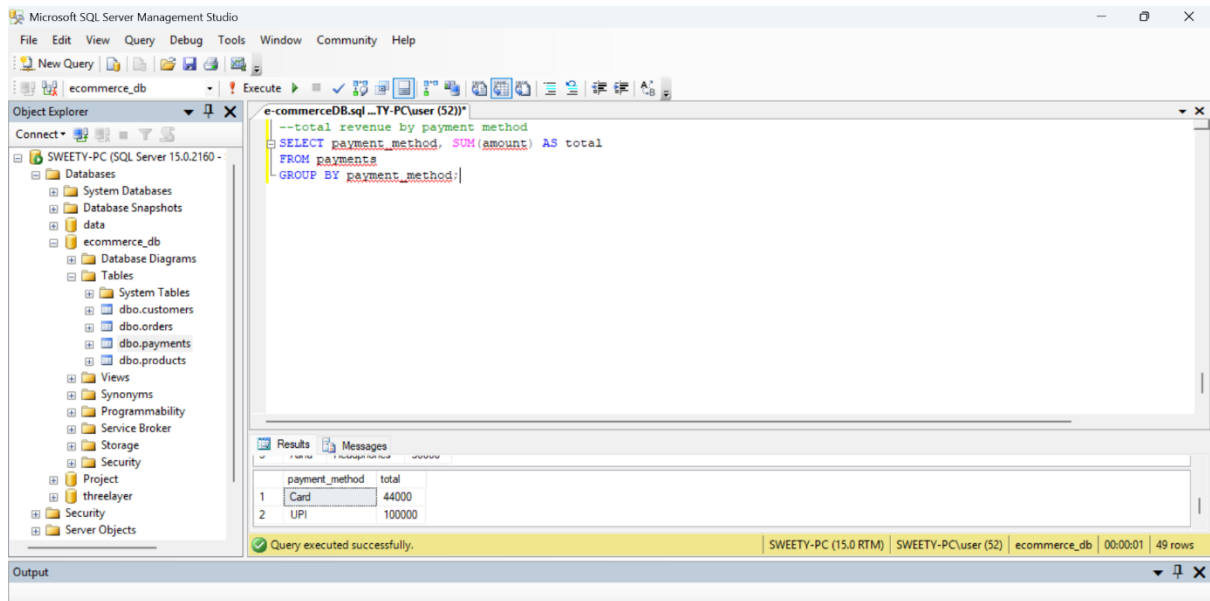
--total revenue by payment method

```
SELECT payment_method, SUM(amount) AS total
```

```
FROM payments
```

```
GROUP BY payment_method;
```

Output:



The screenshot shows the Microsoft SQL Server Management Studio interface. The Object Explorer on the left displays the database structure for 'ecommerce_db', including tables like 'customers', 'orders', 'payments', and 'products'. The central query editor contains the following SQL code:

```
--total revenue by payment method
SELECT payment_method, SUM(amount) AS total
FROM payments
GROUP BY payment_method;
```

The Results pane at the bottom displays the output of the query as a table with two columns: 'payment_method' and 'total'. The data is as follows:

payment_method	total
Card	44000
UPI	100000

The status bar at the bottom indicates that the query was executed successfully, returning 49 rows in 00:00:01.

--find customer who bought laptop

```
SELECT DISTINCT c.name
```

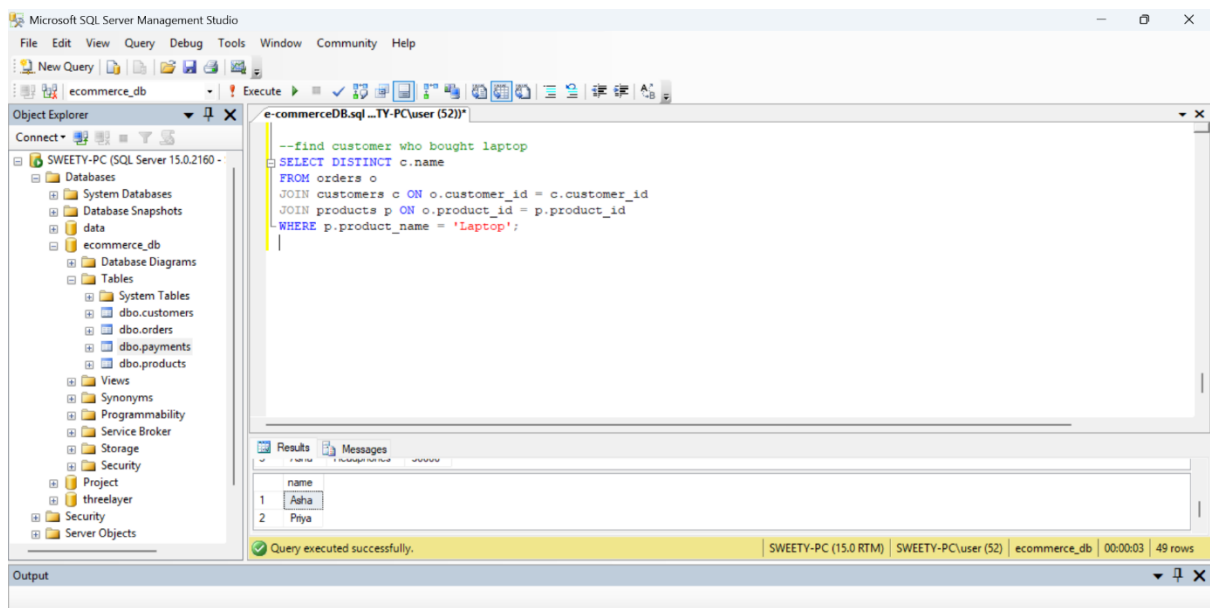
```
FROM orders o
```

```
JOIN customers c ON o.customer_id = c.customer_id
```

```
JOIN products p ON o.product_id = p.product_id
```

```
WHERE p.product_name = 'Laptop';
```

Output:



Conclusion:

This project helped me understand relational database design and SQL query writing. I gained hands-on experience in creating tables, establishing relationships, and analyzing business data using SQL.

This project demonstrates my ability to work with structured data and solve real-world problems using SQL.